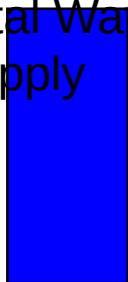


Water Quality

World Water Distribution

Location	% of total
Fresh water lakes	0.009
Saline and inland lakes	0.008
Rivers	0.0001
Soil moisture	0.005
Groundwater (depth above 4k meters)	0.61
Ica caps and glaciers	2.14
Atmosphere	0.001
Oceans	97.3
Total	100

Total Water
Supply



Fresh Water
Supply
Salt Water
Supply

Available
Fresh Water
Supply



Water Pollution

- Definition
 - The presence in water of impurities in such quantity and of such nature as to impair the use of water of a stated purpose.

Water Quality Parameters

- Physical- are those parameters that respond to the senses of sight, touch, or smell. Suspended solids, turbidity, color, taste, temperature and odor fall into this category
- SS in water consist of inorganic and organic particles or of immiscible liquids
- Eg, clay, silt, plant fibres, biological solids (microbes)

Solids – settleable, suspended, dissolved

Three categories:

- settleable
- suspended
- fine or dissolved solids

Rule of Thumb

Solids produced by fish :
0.3 to 0.4 kg TSS for every
1 kg of feed fed

- upper limit: 25 mg TSS/L
- normal operation (species dependent)
 - 10 mg/L for cold water species
 - 20 – 30 mg/L for warm water species

Terms in SS/TDS

- Filterable residues pass through the filter along with the water and relate more closely to dissolved solids while
- Non filterable residues are retained on the filter and retain more closely to suspended solids.

Filterable and non filterable residues are frequently used in laboratory analysis while dissolved and suspended solids are frequently used in water-quality management practice

Total Dissolved Solids

- TDS are the total amount of mobile charged ions, including minerals, salts or metals dissolved in a given volume of water,
- It is expressed in units of mg per unit volume of water (mg/L), also referred to as parts per million (ppm).
- In general, the total dissolved solids concentration is the sum of the cations (positively charged) and anions (negatively charged) ions in the water.
- A TDS meter is based on the electrical conductivity (EC) of water. Pure water has virtually zero conductivity. Typically, the higher the level of EC, the higher the conversion factor to determine the TDS
- Some dissolved solids come from organic sources such as leaves, silt, plankton, and industrial waste and sewage. Other sources come from runoff from urban areas, road salts used on street during the winter, and fertilizers and pesticides used on lawns and farms.
- Dissolved solids also come from inorganic materials such as rocks and air that may contain calcium bicarbonate, nitrogen, iron phosphorous, sulfur, and other minerals.
- Water may also pick up metals such as lead or copper as they travel through pipes used to distribute water to consumers.

TDS standards

- The EPA Secondary Regulations advise a maximum contamination level (MCL) of 500mg/liter (500 parts per million (ppm)) for TDS.
- Numerous water supplies exceed this level.
- When TDS levels exceed 1000mg/L it is generally considered unfit for human consumption. A high level of TDS is an indicator of potential concerns, and warrants further investigation. Most often, high levels of TDS are caused by the presence of potassium, chlorides and sodium. These ions have little or no short-term effects, but toxic ions (lead arsenic, cadmium, nitrate and others) may also be dissolved in the water.

- **Turbidity**- is the measure of extent to which light is either absorbed or scattered material in water, because absorption and scattering are influenced by both size and surface characteristics of the suspended material in water.
- Turbidity meter readings are expressed in Nephelometry Turbidity Units (NTU) or Formazin Turbidity (FTU)(Standard- 1 FTU)
- Turbid water is aesthetically displeasing, and colloidal material associated with turbidity provides adsorption sites for undesirable entities.
- It may also affect photosynthesis of the aquatic plants
- Turbidity provides adsorption sites for chemicals that may be harmful or cause undesirable tastes and odors for biological organisms

COLOUR

- Colour contributed by dissolved solids that remain after removal of suspended matter is known as true color
- Source: organic debris such as leaves, conifer needles, weeds, wood, tannins, humic acid, humates iron oxide cause reddish water, manganese oxides cause brown or blackish water. Industrial waste from textile, chemical production mining, refining, slaughterhouse operations may add substantial color to water.
- Plankton blooms and release of tannic acid from decaying vegetation
- Expressed as true color units where one unit is equivalent to the color produced by 1mg/l of platinum in the form of chloroplatinate ions.

Taste and Odor

- Self definitive terms
- Substances which produces odor will invariably impart the taste to water
- Minerals, metals, salts from soil, end products of biological reactions, and constituents of wastewater
- Alkaline material imparts bitter taste while metallic salts give a salty or bitter taste, organic material (Biological decomposition) produce both taste and odor
- Measurement of taste and odor causing organics can be made using gas or liquid chromatography
- Threshold odor Number (TON) , varying amount of odorous water are poured into containers and diluted with enough odor free distilled water to make a 200 ml mixture.
- An assembled panel of 5-10 “noses” is used to determine the mixture in which the odor is just barely detectable to the sense of smell. The TON of that sample is then calculated using:
- $TON = A+B/A$

- Where A is the volume of odorous water (ml) and B is the volume of odor free water required to produce 200 ml mixture.
- A similar test can be used to quantify taste. Or the panel can simply rate the water quantitatively on an acceptability scale
- A maximum TON of 3 has been recommended by the Public Health Service

Temperature

- It is not used directly to evaluate either potable or waste water. However it is important parameter for surface water and biological species
- Shallow water bodies are more affected as compared to deep water bodies
- An increase of 10 degree C is usually sufficient to double the biological activity if essential nutrients are present, eg algae
- Temperature changes affect the reaction rates and solubility levels of chemicals, dissolution of solids are accelerated by increased temperatures. On the other the solubility of gases decreases at elevated temperature (DO)
- The viscosity of water increase with decreasing temperature. The maximum density is at 4 degree C and density decrease on either side of it.

Three Classifications:

- cold-water species below 15 ° C (60° F)
- cool-water species between 15 °- 20° C (60°- 68° F)
- warm-water species above 20° C (68° F)

Effects of oxygen depletion

- Fish gasp at surface
- Fish group around incoming water
- Fish go off feed
- Nonfish animals leave water
- Slow growth
- Repeated health problems
- Change in water color
- Fish-eating birds present

Causes of O₂ depletion

- Overstocking – aquacrop uses up all the oxygen
- Weather – limited photosynthesis on cloudy days
- Time of day – lowest oxygen levels in hot weather just before sunrise
- Decay of feed – uneaten food will consume oxygen
- Feeding activity – aquacrop uses more oxygen
- Competition – trash fish and pests use oxygen
- Decay of vegetation – dead weeds, leaves, grass uses up oxygen
- Water temperature – the colder the water, the more oxygen it can hold
- Salinity – the saltier the water, the less oxygen it can hold
- Equipment failure – when aeration devices fail

Carbon Dioxide

- Carbon dioxide is a highly soluble in water.
- Concentration in pure water: 0.54 mg/L at 20° C.
- Groundwater concentrations range from 0-100 mg/L.

Exposure to high carbon dioxide concentrations reduces respiration efficiency and decreases the tolerance to low dissolved oxygen concentrations.

Chemical Parameters

- Total Dissolved Solids-The material remaining in the water after filtration for the suspended solids analysis is considered to be dissolved
- This material is left as a solid residue upon evaporation of the water and constitutes a part of total solids
- It results from the solvent action of water on solid, liquid and gases
- Dissolved substances may be organic and inorganic
- Inorganic include mineral, metal and gases
- Dissolved substances are undesirable in water may produce color, tastes, and odors. Some chemicals are toxic and carcinogenic
- TDS is often measured by E_c
- The ability of water to conduct electricity is known as *specific conductance*
- *In cubic centimeter field at 25 degrees and expressed as mS/m*
- Limitation- only ionic substances will be measured, organic substance that dissolve without ionizing are not measured

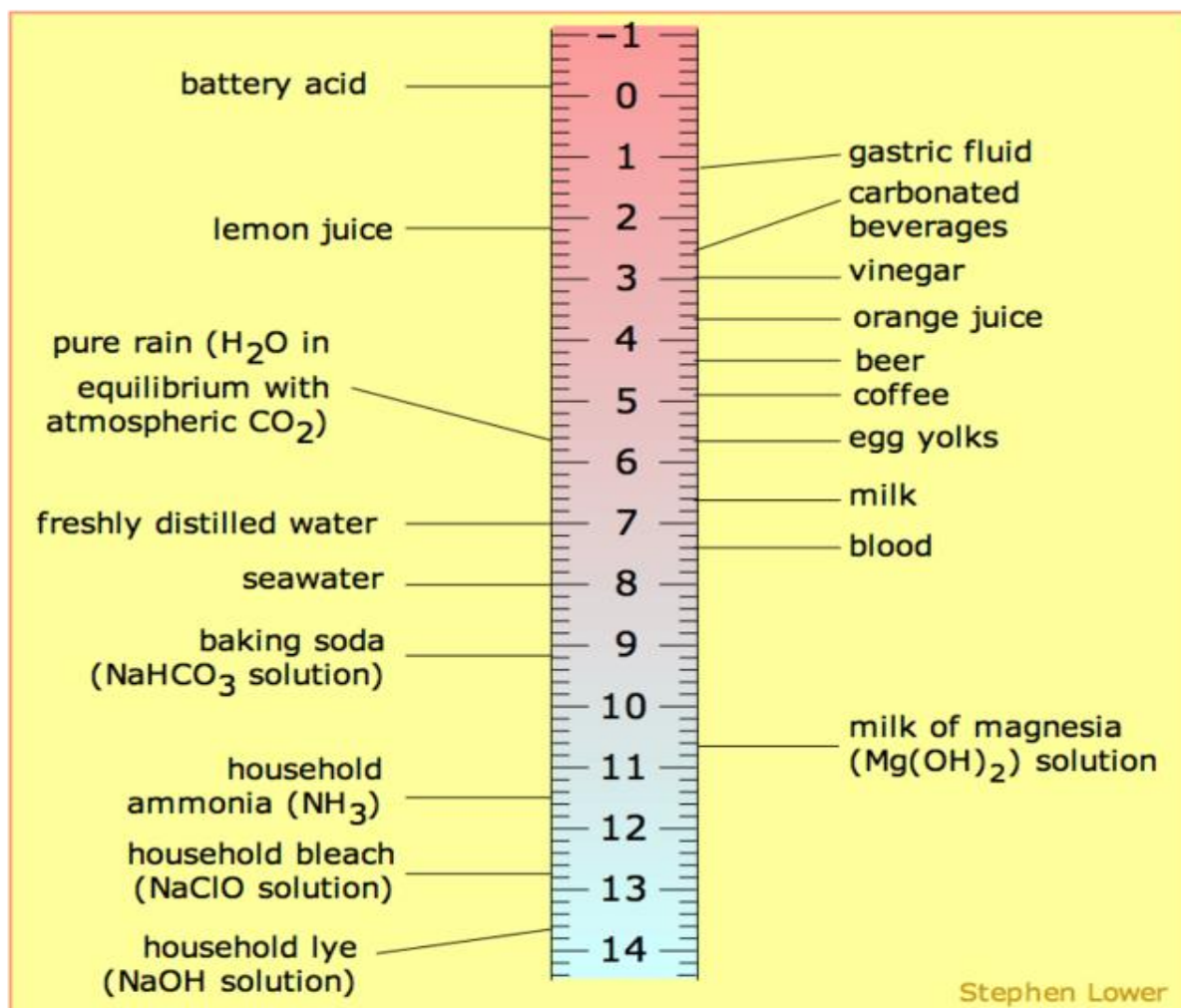
Common Ions in natural waters

Major constituent (1-1000 mg/l)	Secondary constituent (0.01-10mg/l)
Sodium	Iron
Calcium	Strontium
Magnesium	Potassium
Bicarbonate	Carbonate
Sulfate	Nitrate
Chloride	Fluoride, Boron, Silica

Alkalinity

- The quantity of ions in water that will react to neutralise hydrogen ions
- It includes carbonates, bicarbonates, hydroxyl, hydrogensilicates, hydrogenborates, hydrogen phosphates, hydrogen sulphite, ammonia.
- These compounds result from the dissolution of mineral substances in the soil and atmosphere
- Phosphates may also originate from detergents in wastewater and from fertilizers and insecticides
- It imparts a bitter taste to water
- Its high quantity can foul pipes and equipments
- Alkalinity can be measured by titrating with an acid and determining hydrogen equivalent

pH Scale



Hardness

- The concentration of multivalent metallic cations in solution
- It is classified as carbonate hardness and noncarbonate hardness
- The hardness that is equivalent to alkalinity is termed as carbonate hardness and remaining is noncarbonate
- The multivalent metallic ions are abundant in natural water are calcium, magnesium, iron, manganese, strontium, aluminium
- Soap consumption by hard waters represents an economic loss to water user
- Sodium soaps react with multivalent cations to form a precipitate, thereby losing their surfactant property
- Scaling in boilers
- Limit 500mg/l

IMPACTS

- Leathering does not occur until all the hardness ions are precipitated, at which point the water has been softened by the soap.
- The precipitate formed by hardness and soap adheres to surfaces of tubs, sinks, dishwashers and may stain clothing, dishes and other items.
- Residues of the hardness-soap precipitate may remain in the pores, so that skin may feel rough and uncomfortable.
- In recent years these problems have been largely alleviated by the development of soaps and detergents that do not react with hardness.

- Boiler Scale, the result of carbonate hardness precipitate may cause considerable economic loss through fouling of water heaters and hot waer pipes.
- Changes in pH in water distribution system may also result in deposits of precipitates.
- Bicarbonates begin to convert to the less soluble carbonates at pH above 9
- Magnesium hardness associated with sulfate ion has a laxative effect.
- Calcium hardness present no health problem, infact hard water is apparantly beneficial to the human cardiovascular system.

Hardness

Classified as:

soft (0-75 mg/L)

moderately hard (75 – 150 mg/L)

hard (150-300 mg/L)

very hard (> 300 mg/L)

Recommended range: 20 to 300 mg/L CaCO_3

Fluoride

- Associated with sedimentary and igneous rocks
- Seldom in surface water, and appear in GW in few geographical locations
- F is toxic in large quantities
- 1mg/l is useful as form stronger teeth in children, while high quantity result in discoloration and mottling, also cause other skeleton abnormalities

- Fluorides are compounds that combine the element fluorine with another substance, usually a metal. Examples include sodium fluoride, stannous fluoride, and fluoride monofluorophosphate
- Fluorides occur naturally in soil, air, or water.
- The types of fluoride added to different water systems include fluorosilicic acid, sodium fluorosilicate, and sodium fluoride.
- The US Environmental Protection Agency (EPA) has set a maximum amount of fluoride allowable in drinking water of 1.0 mg/L.
- BIS set a maximum limit of 1.5 mg/l
- Long-term exposure to levels higher than this can cause a condition called *skeletal fluorosis*, in which fluoride accumulates in the bones. This can eventually result in joint stiffness and pain, and can lead to weak or brittle bones in older adults

Fluoride removal

- Absorption on Activated Granular Alumina (pH 5,0 - 6,0) and following washing and regeneration
- Reverse Osmosis
- ED and EDR (Reverse Electrodialysis)
- Absorption on Bone Char
- These techniques, applicable for the elimination of Fluorides in low concentrations
- Companies and Organizations have studied systems of precipitation of the Fluoride ion in sedimentation plants, suitable for small and medium flow dimension waste water production.
- A wide literature is available on methods of precipitation of the Fluorides with salts of Calcium, Aluminum and Iron, that exploit the lower Products of Solubility of the correspondents fluorinated salts.
- The treatment of water containing Fluorides, has been traditionally realized with Lime in alkaline pH range.
 - The reaction is :
 - $\text{Ca(OH)}_2 + 2 \text{HF} = \text{CaF}_2 + \text{H}_2\text{O}$

Water Quality Parameters

Critical Parameters

- Dissolved Oxygen
- Temperature
- Ammonia/Nitrite/Nitrate
- pH

Important Parameters

- Alkalinity/Hardness
- Salinity
- Carbon Dioxide
- Solids

Metals

- All metals are soluble to some extent
- Metals posing health hazard in relatively less amount are labeled toxic
- These occur in natural water through natural deposits, discharges of domestic, industrial and agricultural waste.
- Measured by AAS
- Non toxic metals – Na, Fe, Mn, Al, Cu, Zn.
- Fe and Mn occur together and cause no hazard at natural conc., however, Fe (0.3mg/l) and Mn (0.05mg/l) can cause color problems and become source of energy for bacteria resulting in slime growth.
- Fe usually found as FeCl_2 , $\text{Fe}(\text{HCO}_3)_2$, FeSO_4 . Thus significant Fe is found only in systems devoid of Oxygen such as GW and lower layer of lakes.
- Mn is associated with chloride, nitrate and sulfates.
- Cu and Zn are synergistic and may be toxic even in small quantities.
- Excessive conc. Add bitter taste to water and health hazard to cardiac and kidney patients

- Toxic metal- As, Ba, Cd, Cr, Pb, Hg, Ag
- These metals concentrate in food chain
- In natural systems these are found in minute quantities.
- Significant conc. In water is usually caused by mining, industrial and agricultural sources

Heavy Metals in water

- Heavy metals are elements having atomic weights between 63.546 and 200.590, and a specific gravity greater than 4.0 Living organisms require trace amounts of some heavy metals, including cobalt, copper, iron, manganese, molybdenum, vanadium, strontium, and zinc.
- All heavy metals exist in surface waters in colloidal, particulate, and dissolved phases
- The colloidal and particulate metal may be found in 1) hydroxides, oxides, silicates, or sulfides; or 2) adsorbed to clay, silica, or organic matter
- The solubility of trace metals in surface waters is predominately controlled by the water pH, the type and concentration of ligands on which the metal could adsorb, and the oxidation state of the mineral components and the redox environment of the system

Source

- *Nonpoint sources:*
 - Natural: Chemical and physical weathering of igneous and metamorphic rocks and soils often release heavy metals into the sediment and into the air. Other contributions include the decomposition of plant and animal detritus, precipitation or atmospheric deposition of airborne particles from volcanic activity, wind erosion, forest fire smoke, plant exudates, and oceanic spray (Kennish, 1992).
 - Anthropogenic: Surface runoff from mining operations usually has a low pH and contains high levels of metals such as iron, manganese, zinc, copper, nickel and cobalt. The combustion of fossil fuels pollutes the atmosphere with metal particulates that eventually settle to the land surface. Currently, anthropogenic inputs of metals exceed natural inputs.
- *Point sources:* Domestic wastewater effluent contains metals from metabolic wastes, corrosion of water pipes, and consumer products. Industrial effluents and waste sludges may substantially contribute to metal loading

H.M. combinations

- The behavior of metals in natural waters is a function of the substrate sediment composition, the suspended sediment composition, and the water chemistry.
- Sediment composed of fine sand and silt will generally have higher levels of adsorbed metal than will quartz, feldspar, and detrital carbonate-rich sediment.
- Metals also have a high affinity for humic acids, organo-clays, and oxides coated with organic matter.

Conti...

- Heavy metals like lead, mercury, iron, cadmium, aluminum and magnesium are present in water sources. If these metals are present in the sediment, these reach the food chain through plants and aquatic animals. This causes heavy metal poisoning in case the level in the water is very high.
- In older smelters, air and water emissions contained elevated levels of various metals. Copper and selenium, for example, which can be released from copper smelters, are essential to organisms as trace elements, but they are toxic if they are overabundant. These metals can **contaminate** the soil in the vicinity of smelters, destroying much of the vegetation.

Health Effects

- Oxides of such toxic metals as arsenic (cumulative poison), cadmium (heart disease), and mercury (nerve damage).
- Blood diseases, nervous system disorders and heart diseases are some of the effects of water pollution. Rarely, the body's chromosomal makeup can be altered. Some of the less potent effects are skin lesions, vomiting and diarrhea
- Ingestion of metals such as lead (Pb), cadmium (Cd), mercury (Hg), arsenic (As), barium (Ba), and chromium (Cr), may pose great risks to human health.
- Trace metals such as lead and cadmium will interfere with essential nutrients of similar appearance, such as calcium (Ca^{2+}) and zinc (Zn^{2+}).

- *Lead:* Because of size and charge similarities, lead can substitute for calcium and included in bone. Children are especially susceptible to lead because developing skeletal systems require high calcium levels. Once free in the system, lead may cause nephrotoxicity, neurotoxicity, and hypertension.
- *Cadmium:* Cadmium may interfere with the metallothionein's ability to regulate zinc and copper concentrations in the body. Metallothionein is a protein that binds to excess essential metals to render them unavailable. When cadmium induces metallothionein activity, it binds to copper and zinc, disrupting the homeostasis levels. Cadmium is used in industrial manufacturer and is a byproduct of the metallurgy of zinc.

Conti...

- *Mercury*: Mercury poses a great risk to humans, especially in the form of methylmercury. When mercury enters water it is often transformed by microorganisms into the toxic methyl mercury form. Symptoms of acute poisoning are pharyngitis, gastroenteritis, vomiting, nephritis, hepatitis, and circulatory collapse. Chronic poisoning is usually a result of industrial exposure or a diet consisting of contaminated fish (mercury is the only metal that will bioaccumulate). Chronic poisoning may cause liver damage, neural damage, and teratogenesis
- *Arsenic*: Arsenic ingestion can cause severe toxicity through ingestion of contaminated food and water. Ingestion causes vomiting, diarrhea, and cardiac abnormalities
- *Chromium*: The presence of abundant chromium anions in the water is generally a result of industrial waste. The chronic adverse health effects are respiratory and dermatologic

conti...

- **Irrigation Effects:** Irrigation water may transport dissolved heavy metals to agricultural fields. Although most heavy metal do not pose a threat to humans through crop consumption, cadmium may be incorporated into plant tissue. Accumulation usually occurs in plant roots, but may also occur throughout the plant

Arsenic Pollution

- Arsenic is a widely dispersed element in the Earth's crust and exists at an average concentration of approximately 5 mg/kg.
- The introduction of arsenic into drinking water can occur as a result of its natural geological presence in local bedrock. Commonly found in Bangladesh, West Bengal (India), and regions of China,
- Natural contamination of surface waters and soil can arise when arsenic-rich geothermal fluids come into contact with surface waters
- Groundwater contamination by arsenic arises from sources of arsenopyrite, base metal sulfides, realgar and orpiment, arsenic-rich pyrite, and iron oxyhydroxide
- Arsenic enrichment also takes place in geothermally active areas; surface waters are more susceptible than groundwater to contamination in the vicinity of such geothermal systems, and evidence suggests that increased use of geothermal power may elevate risks of arsenic exposure in affected areas

A recent study which finds that **arsenic pollution in groundwater is caused by the indiscriminate use of chemicals in agriculture**



Organic Pollutants

- Natural organics consist of decay products of organic solid
- Synthetic organic result from wastewater discharge or agricultural practices
- Dissolved organics are categorized as:
 1. Biodegradable
 2. Non biodegradable (refractory)

Biodegradable Organics

- Consist of organics that can be utilized for food by naturally occurring microbes
- Mainly starch, fats, proteins, alcohols, acids, aldehyde, esters.
- They may be end products of initial microbial decomposition of plant and animal tissues. Or result of from domestic and industrial wastewater discharges.
- Main problem is color, smell and taste
- Secondary problems arises due to microbial decomposition.
- Microbial decomposition can be accompanied by oxidation and reduction process.

Aerobic/Anaerobic decomposition

- in aerobic environment the end products of microbial decomposition are stable and acceptable compounds.
- Anaerobic decomposition results in unstable and objectionable end products.
- Measured by the process of BOD estimation

(Biological Oxygen Demand) BOD

- The amount of oxygen required by microbes for biochemical decomposition process is called BOD
- Commonly used is assessing water quality
- 5 day BOD
- The so-called 5-day BOD measures the amount of oxygen consumed by biochemical oxidation of waste contaminants in a 5-day period. The total amount of oxygen consumed when the biochemical reaction is allowed to proceed to completion is called the Ultimate BOD. The Ultimate BOD is too time consuming, so the 5-day BOD has almost universally been adopted as a measure of relative pollution effect.
- When BOD is high then DO become too low to support life in water
- The US EPA defines the threshold for a water pollution alert as a DO content of less than 5 mg/l.
- The presence of toxic substances in water will invalidate the BOD results.

Non biodegradable Organics

- Materials resistant to biological degradation
- Tannic acid, cellulose, phenols, constituents of woody plants
- Molecules with exceptionally strong bonds
- And ringed structures, pesticides, insecticides, herbicides, and detergents like Alkyl benzene sulfonate (ABS)- does not biodegrade.
- Measured by the process of COD estimation, also by TOC analysis.
- Both COD and TOC measure the biodegradable fraction of the organics, so BOD must be subtracted from COD/TOC values to quantify nonbiodegradable organics
- Specific organic compounds can also be quantified and identified by gas chromatography

Chemical Oxygen Demand (COD)

- The BOD test measures the oxygen demand of biodegradable pollutants whereas the COD test measures the oxygen demand of biodegradable pollutants plus the oxygen demand of non-biodegradable oxidizable pollutants.
- Any oxidizable material present in a natural waterway or in an industrial wastewater will be oxidized both by biochemical (bacterial) or chemical processes. The result is that the oxygen content of the water will be decreased. Basically, the reaction for biochemical oxidation may be written as:
- **Oxidizable material + bacteria + nutrient + O₂ ==> CO₂ + H₂O + oxidized inorganics such as NO₃ or SO₄**
Both the BOD and COD tests are a measure of the relative oxygen-depletion effect of a waste contaminant. Both have been widely adopted as a measure of pollution effect.

Nutrients (N/P)

- Nutrients required in abundance by aquatic species is C, N, And P
- C- readily available from atm., alkalinity, and decay products of Organic matter.
- Nitrogen- part of amino acids- proteins
- Upon decomposition, protein breakdown to amino acids-ammonia-nitrite (in presence of oxygen)- nitrate- reconstituted to organic matter
- Secondly from animal wastes, chemical fertilizers and wastewater discharges.
- Nitrogen from these sources may be discharged directly into streams or may enter water ways through surface runoff or groundwater discharge.
- Nitrogen compounds can be oxidized to nitrate by bacteria and may be carried to GW by percolating water.
- Once in aquifer, nitrates move freely with the ground water flow.
- Nitrogen can have other severe consequences in form of ammonia at temperatures and pressures normally found in natural water systems.

- Nitrate poisoning in infant animals and human cause serious problem and even death.
- Low acidity in an infant's intestinal tract permits growth of nitrate reducing bacteria that convert the nitrate to nitrite which is then absorbed into the blood stream
- Nitrite has great affinity for haemoglobin than does oxygen and thus replace it., this suffocates the victim.
- Oxygen starvation result in a bluish coloration of the body , nitrate poisoning has been referred to as the blue baby syndrome, also k/as *methemoglobinemia*

- Once the flora of the intestinal tract has fully developed, usually after 6 months, nitrate conversion to nitrite is seldom a problem.
- Fortunately, the natural oxidation of nitrite to nitrate occurs quickly so that significant quantities of nitrites are not found in natural water.
- Tests for nitrogen forms in water commonly include analysis for Ammonia (NH_3 , NH_4^+), nitrate, Organic N
- Tests for Ammonia and organic N are for wastewater while test for nitrate is most common for clean water samples

- Phosphorous appears as Phosphates in aquatic environment.
- Other forms include orthophosphates, condensed Phosphates (pyro, meta, polyphosphates) and organically bound Phosphates.
- These may be in soluble or particulate form or may be constituents of plant or animal tissue.
- Like nitrogen phosphates pass through the cycles of decomposition and photosynthesis.



- Phosphate is constituent of soil and is used extensively in fertilizer to supplement agricultural lands, it is also a constituent of animal waste, municipal wastewater and industrial waste
- Phosphate are not toxic but do represent a serious indirect threat to water quality.
- Its available quantity result in rapid growth of aquatic plants leading to severe consequences.
- The tendency of phosphate to adsorb to soil particles limits its movement in soil moisture and GW., but results in its transport into surface waters by erosion.

- Municipal wastewater is another major source of P in surface water.
- Condensed P are used extensively in detergents and organic P are constituents of body waste and food residue.
- Other sources include industrial waste in which phosphate compounds are used for boiler water conditioning.
- P are not toxic, and no direct health effects, but do represent a serious indirect threat to water quality.
- They can also interfere with wastewater treatment processes.
- Measured colorimetrically

Biological water Quality parameters

- Pathogens
 - Bacteria, Virus, Protozoa, helminths

Bacteria (cocci, bacilli)

Cholera (*Vibrio Comma*),

Typhoid(*Salmonella typhosa*),

Viruses

Enteric cytopathogeinc human orphan , Infectious hepatitis

Protozoa

Entamoeba histolytica

Helminths

Parasitic worms (tape worms, bood fluke)

Bioindicators

- Bioindicators include biological processes, species, or communities and are used to assess the quality of the environment and how it changes over time.
- Changes in the environment are often attributed to anthropogenic disturbances (e.g., pollution, land use changes) or natural stressors (e.g., drought, late spring freeze),
- Species or group of species of known environmental requirements that respond to alterations of their habitats with changes in abundance, physiology and behavior

The ideal Bioindicator

- Easy taxonomical identification
- World wide distribution
- Numerical abundancy
- Low genetic, ecological variability
- Long life cycle
- Known ecological characteristics
- Easy availability in laboratory

Why Bioindicators

- To monitor the changes in environment
- For e.g. to evaluate the state of water body and variability of its quality
- To establish actions to maintain its quality
- To evaluate the efficiency of adopted actions

- the direct determination of biological effects,
- the determination of synergetic and antagonistic effects of multiple pollutants on an organism,
- the early recognition of pollutant damage to plants as well as toxic dangers to humans and
- relatively low cost compared to technical measuring methods

Categorized as:

- **Accumulation indicators** store pollutants without any evident changes in their metabolisms.
- **Response indicators** react with cell changes or visible symptoms of damage when taking up even small amounts of harmful substances.